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BRIEF COMMUNICATION

The effect of levonorgestrel-releasing intrauterine devices on acne vulgaris: A prospective cohort study

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KEYWORDS

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The levonorgestrel-releasing intrauterine device (LNG-IUD) may be associated with worsening acne vulgaris due to the second-generation progestin's androgenic effects.[1–3] Combined oral contraceptives (COCs) have a therapeutic effect because ethinylestradiol increases sex-hormone binding globulin and suppresses ovulation. These processes reduce circulating free androgens and decrease androgen production, thus decreasing sebaceous gland activity and acne.[2] Acne has been cited as the second most common reason after bleeding irregularities for premature IUD removal.[4] Given increasing IUD use, studying potential after effects may improve patient satisfaction. Current literature does not delineate the effects of patients' prior birth control methods (BCMs) on acne development. The aim of the present study was to determine whether the observed worsening of acne severity post-LNG-IUD placement is due to LNG-IUD hormonal effects or withdrawal from COCs.

This prospective cohort study was approved by the Colorado Multiple Institutional Review Board (protocol no.: 17-1903) and included participants ≥18 years of age that had an LNG-IUD (52 mg) placed at a University of Colorado Health Obstetrics and Gynecology or family medicine clinic. Exclusion criteria included women <12 weeks postpartum or acne treatment changes within 12 weeks before IUD placement. Progression of acne was rated at 0-, 12-, and 24-weeks post-placement using facial photographs from three angles (central, right, left). Two board-certified dermatologists independently rated acne severity on a scale of 0 (clear) to 5 (very severe) using the Evaluator's Global Acne Severity Scale (GASS).[5] A total of 21 participants were enrolled, and patient-reported BCM used within 10 days of IUD placement was recorded (Table S1). The non-COC BCM group included participants who had used progestin-only pills, barrier methods, or had no BCM prior to IUD placement. Three participants dropped out of the study due to loss to follow-up (1), change in acne treatment after IUD placement (1), and removal of IUD due to other side effects (1). GASS ratings were compared between weeks 0, 12, and 24 using intention-to-treat analyses and paired student's *t*-tests. Patients who did not complete the study received scores of "0" for subsequent timepoints to include their week 0 scores in paired *t*-test analyses. Unpaired student's *t*-tests were used to compare GASS means between COC and non-COC groups at each timepoint (GraphPad Prism 9.5.1, San Diego, California, USA).

Acne severity worsened overall at 24-weeks post-LNG-IUD placement (GASS +0.07) with six participants scoring at least an average GASS of 2 (mild) by week 24 (Table S1). When stratified by

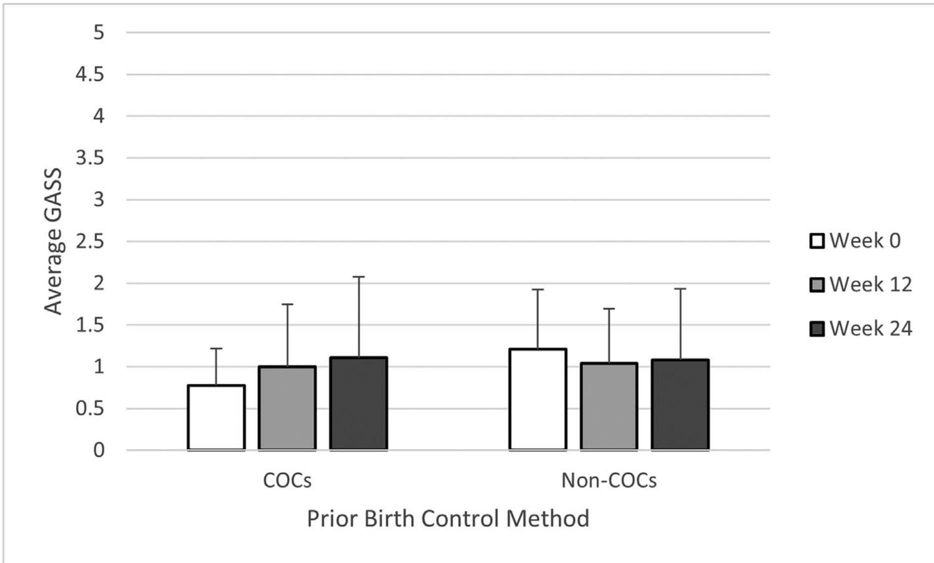

FIGURE 1 Mean acne scores at 0-, 12-, and 24-weeks post-levonorgestrel-releasing intrauterine device (LNG-IUD) placement stratified by prior birth control method (BCM).

prior BCM, a numerical increase in mean acne score at 12- (+0.22, 95% confidence interval [CI]; −0.33 to 0.77, $P=0.38$) and 24-weeks (+0.33, 95% CI: −0.36 to 1.03, $P=0.30$) post-LNG-IUD placement was observed in the COCs group that was not observed in the non-COCs group (Figure 1). There was no significant difference in mean GASS between the COC and non-COC groups at 0 weeks ($P=0.13$).

This exploratory study suggests withdrawal of acne-protective COCs may be associated with worsened acne severity post-LNG-IUD placement compared to the hormonal side effects of the LNG-IUD itself, which is consistent with prior studies.[1] In our stratified analysis, participants who switched from COCs to the LNG-IUD had a numerical increase in mean acne score over the 24-week period compared to participants who switched from a non-COC. Although our study found the mean increase in GASS to be numerically small over the study period, changes in acne may be significant for patients who, prior to IUD placement, were almost clear of acne and were now beginning to develop acne after IUD placement. Additionally, one patient in our cohort dropped out of the study due to worsening acne and needing to change their acne treatment, illustrating that changes in acne occur at an individual level and may be clinically significant for a patient. Our study found no significant increase in acne score for participants who were on non-COC BCMs prior to IUD placement.

Limitations of this study include low participant number potentially contributing to imprecision of results, lack of evaluation of acne on the chest and back, and lack of assessing formal lesion counts and patient-reported outcomes. The non-COC group also included both hormonal and non-hormonal BCMs, and this heterogeneity may obscure method-specific effects of individual BCMs. Strengths of the study include the prospective nature. Both COC withdrawal and IUD placement introduce hormonal changes that may contribute to acne, and while this preliminary study did not definitively isolate hormonal effects, future large-scale clinical studies are warranted to further examine how prior BCM plays a role in acne pathogenesis.

AUTHOR CONTRIBUTIONS

Drs Stevenson, Goodman, Wang, Hugh, Guidry, Mirhossaini, Newman, and Stephen contributed substantially to the conception and design of the study. Drs Cao, Zita, and Kim contributed substantially to the acquisition, analysis, and interpretation of data for the work. All authors contributed to drafting the work and/or reviewing it critically for important intellectual content. All authors gave their final approval of the version to be published. All authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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CONFLICT OF INTEREST STATEMENT

Dr. Lazorwitz is Chief Medical Advisor for Dama Health and a Scientific Advisory Board member for 3Daughters. Drs Cao, Zita, Kim, Stevenson, Goodman, Wang, Hugh, Guidry, Mirhossaini, Newman, and Stephen have no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data available on request from the authors. The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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